



ABBOTSLEIGH

FILE

AUGUST 2006

YEAR 12

ASSESSMENT 4

TRIAL HIGHER SCHOOL CERTIFICATE  
EXAMINATION

# Mathematics Extension 1

## General Instructions

- Reading time – 5 minutes.
- Working time – 2 hours.
- Write using blue or black pen.
- Board-approved calculators may be used
- A table of standard integrals is provided with this paper
- All necessary working should be shown in every question

## Total marks - 84

- Attempt Questions 1-7
- All questions are of equal value
- Answer each question in a separate writing booklet

Total marks – 84  
Attempt Questions 1–7  
All questions are of equal value

Answer each question in a SEPARATE writing booklet. Extra booklets are available.

Question 1 (12 marks) Use a SEPARATE writing booklet.

Marks

(a) Differentiate  $y = \sin^{-1}(x^2)$ .

2

(b) Use the substitution  $u = 4 - x$  to evaluate  $\int_0^3 \frac{x}{\sqrt{4-x}} dx$ .

3

(c) Evaluate  $\int_1^{\sqrt{2}} \frac{dx}{\sqrt{2-x^2}}$ .

2

(d) Find  $\int \sin^2 3x dx$ .

2

(e) Solve  $\frac{1}{x+2} \leq 2$

3

Question 2 (12 marks) Use a SEPARATE writing booklet.

Marks

(a) Solve the equation  $\sin 2\theta = 2\cos^2 \theta$  for  $0 \leq \theta \leq 2\pi$ .

3

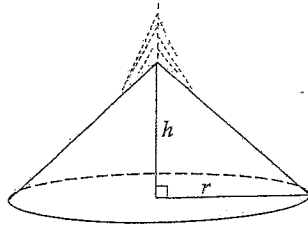
(b) (i) Express  $\sin x + \cos x$  in the form  $R \sin(x + \alpha)$ .

2

(ii) Hence, or otherwise, sketch the graph of  $y = \sin x + \cos x$ , showing the endpoints, in the domain  $0 \leq \theta \leq 2\pi$ .

2

(c) Wheat runs from a hole in a silo at a constant rate and forms a conical heap whose base radius,  $r$ , is three times its height,  $h$ .



NOT TO SCALE

After 1 minute, the height of the heap is 20 cm. Find:

(i) The volume of the conical heap after 1 minute. Leave your answer in terms of  $\pi$ .

2

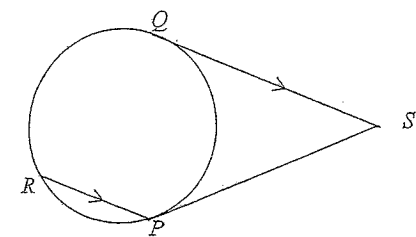
(ii) The rate at which the height of the heap is rising after 1 minute.

3

Question 3 (12 marks) Use a SEPARATE writing booklet.

Marks

(a)



NOT TO SCALE

3

$P$  and  $Q$  are points on a circle and the tangents to the circle at  $P$  and  $Q$  meet at  $S$ .  $R$  is a point on the circle so that  $RP$  is parallel to  $QS$ .

Copy or trace the diagram into your writing booklet.

Prove that  $QP = QR$ .

(b) (i) Find the value of  $k$  if  $2x + 1$  is a factor of  $P(x) = 2x^3 - x^2 + kx + 1$ .

2

(ii) Show that when  $k$  has this value,  $P(x)$  has only the one real root,  $x = -\frac{1}{2}$ .

3

(c) The rate at which a body cools in air is proportional to the difference between its temperature,  $T$  and the constant temperature,  $P$  of the surrounding air. This rate can be expressed by

$$\frac{dT}{dt} = k(T - P)$$

(i) Show that  $T = P + Ae^{kt}$  where  $A$  and  $k$  are constants, satisfies this equation.

1

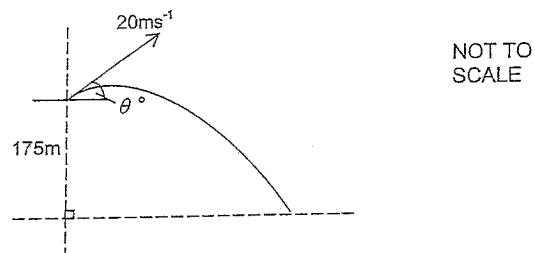
(ii) A heated body cools from  $40^\circ$  to  $30^\circ$  in 1 hour. The air temperature around the body is  $20^\circ$  C. Find the temperature of the body after a further 2 hours.

3

Question 4 (12 marks) Use a SEPARATE writing booklet.

Marks

- (a) When  $(3+x)^n$  is written as a polynomial in  $x$ , the coefficient of  $x^4$  is twice the coefficient of  $x^3$ . Find the value of  $n$ .
- (b) A man standing on top of a vertical cliff throws a stone into the air at an angle  $\theta$  to the horizontal. The top of the vertical cliff is 175 metres above a flat sea.



The angle of projection to the horizontal is  $\theta$ , as shown.  
Let  $(x, y)$  be the position of the stone at time  $t$  seconds after being thrown.  
The initial velocity of the stone is  $20 \text{ ms}^{-1}$ .

You may assume that the path of the stone where the acceleration due to gravity is  $-10 \text{ ms}^{-2}$ , is given by the parametric equations below

$$x = 20t \cos \theta$$

$$y = 20t \sin \theta - 5t^2 + 175$$

(Do NOT prove these equations.)

The angle of projection of the bullet to the horizontal,  $\theta$  is  $30^\circ$ .

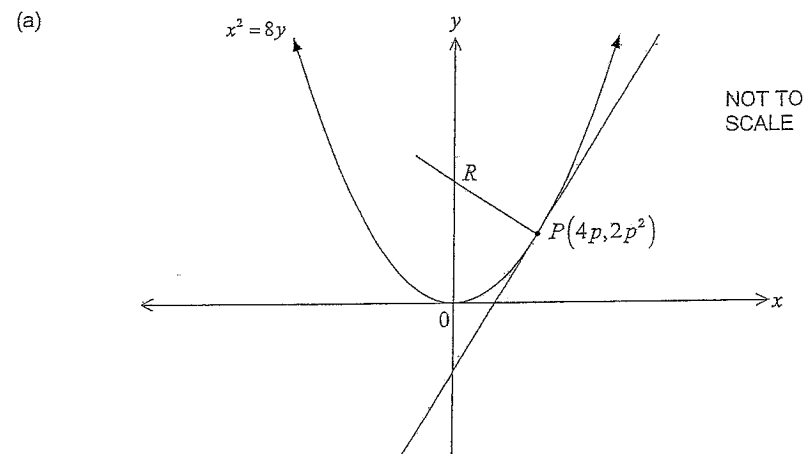
- (i) Find the time it takes for the stone to hit the water.
- (ii) Find the speed at which the stone hits the water.
- (c) A particle moves on a line so that its distance from the origin at time  $t$  seconds is  $x$  cm. Its acceleration is given by

$$\frac{d^2x}{dt^2} = 10x - 2x^3$$

- (i) If its velocity is  $v$  and the particle changes direction 1 cm to the right of the origin, find  $v^2$  in terms of  $x$ .
- (ii) Explain why the particle can never reach the origin.

Question 5 (12 marks) Use a SEPARATE writing booklet.

Marks



Consider the variable point  $P(4p, 2p^2)$  on the parabola  $x^2 = 8y$ .

- (i) Prove that the equation of the normal at  $P$  is  $x + py = 2p^3 + 4p$ .
- (ii) Find the coordinates of the point  $R$ , where the normal at  $P$  intersects the  $y$ -axis.
- (iii) If  $M$  is the midpoint of  $PR$ , find the equation of the locus of  $M$  in Cartesian form.

- (b) Use mathematical induction to prove that for all integers  $n = 1, 2, 3, \dots$

$$1 \times 1! + 2 \times 2! + 3 \times 3! + \dots + n \times n! = (n+1)! - 1$$

- (c) The three roots of the polynomial equation  $x^3 - 3px^2 + 2qx - r = 0$  form an arithmetic series.

Prove that  $r = 2p(q - p^2)$ .

Question 6 (12 marks) Use a SEPARATE writing booklet.

Marks

(a) (i) Sketch the graph of the function  $f(x) = x^3 - 1$  in the domain  $0 \leq x \leq 2$ , clearly showing the coordinates of any points of intersection with the axes. 1

(ii) On the same diagram sketch the graph of the inverse function  $f^{-1}(x)$ , showing the coordinates of any points of intersection with the axes. 2

(iii) Explain why the  $x$ -coordinate of any point of intersection of the graphs  $y = f(x)$  and  $y = f^{-1}(x)$  satisfies the equation  $x^3 - x - 1 = 0$ . 1

(iv) Show that the equation  $x^3 - x - 1 = 0$  has a root between  $x = 1$  and  $x = 2$ . 2

(v) Taking  $x = 1.5$  as the first approximation to the root, use one application of Newton's method to find a better approximation to the root, correct to 3 significant figures. 3

(b) The coefficient of  $x^k$  in  $(1+x)^n$  where  $n$  is a positive integer is denoted as  $\binom{n}{k}$ .

Prove that  $\binom{n}{1} + 2\binom{n}{2} + 3\binom{n}{3} + \dots + (n-1)\binom{n}{n-1} = n(2^{n-1} - 1)$  3

Question 7 (12 marks) Use a SEPARATE writing booklet.

Marks

(a) Solve the equation  $(n + \frac{1}{n})^2 - 5(n + \frac{1}{n}) + 6 = 0$ . 2

(b) (i) Show that the function  $y = \frac{e^x - e^{-x}}{e^x + e^{-x}}$  has no stationary points. 3

(ii) Prove that the lines  $y = \pm 1$  are asymptotes. 2

(iii) Sketch the curve. 1

(iv) If  $k$  is a positive constant, show that the area in the first quadrant enclosed by the above curve and the lines  $y = 1$ ,  $x = 0$  and  $x = k$  is given by

$$\text{Area} = k - \ln(e^k + e^{-k}) + \ln 2$$

(v) Prove that for all positive values of  $k$ , this area is always less than  $\log_e 2$ . 2

END OF PAPER

QUESTION 1

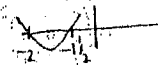
a)  $\frac{dy}{dx} = \frac{2x}{\sqrt{1-x^2}}$   
 $= \frac{2x}{\sqrt{1-x^2}}$

b) Let  $u = 4-x$   $\therefore du = -dx$   
 $x=3, u=1$   
 $x=0, u=4$

$\therefore I = \int_4^1 \frac{4-u}{\sqrt{u}} x - du$   
 $= \int_1^4 \frac{4-u}{\sqrt{u}} du$   
 $= \int_1^4 (4u^{-1/2} - u^{1/2}) du$   
 $= [2x^4 u^{1/2} - \frac{2}{3} u^{3/2}]_1^4$   
 $= 8 \times \sqrt{4} - \frac{2}{3} \times 4 \times \sqrt{4} - (8 - \frac{2}{3})$   
 $= 16 - \frac{16}{3} - 8 + \frac{2}{3} = 3\frac{1}{3}$

c)  $\int \frac{dx}{\sqrt{1-x^2}}$   
 $= [\sin^{-1} \frac{x}{\sqrt{2}}]_{-\frac{\sqrt{2}}{2}}^{\frac{\sqrt{2}}{2}}$   
 $= \sin^{-1} \frac{\sqrt{2}}{\sqrt{2}} - \sin^{-1} \frac{-1}{\sqrt{2}}$   
 $= \frac{\pi}{2} - \frac{\pi}{4}$   
 $= \frac{\pi}{4}$

d)  $I = \int \frac{1}{2} (1 - \cos 6x) dx$   
 $= \frac{1}{2} [x - \frac{\sin 6x}{6}] + C$   
 $= \frac{x}{2} - \frac{\sin 6x}{12} + C$

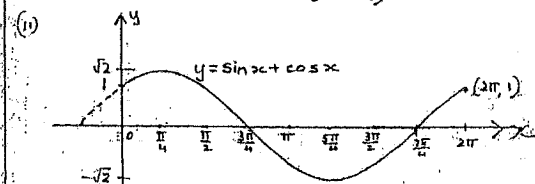
e)  $\frac{1}{(x+2)} \leq 2$   $x \neq -2$ , x b.s. by  $(x+2)^2$   
 $(x+2) \leq 2(x+2)^2$   
 $(x+2)(2(x+2)-1) \geq 0$   
 $(x+2)(2x+3) \geq 0$   
  
 $x < -2$  or  $x > -1\frac{1}{2}$

QUESTION 2

a)  $\sin 2\theta = 2 \cos^2 \theta$   
 $2 \sin \theta \cos \theta = 2 \cos^2 \theta$   
 $\cos \theta (\sin \theta - \cos \theta) = 0$   
 $\cos \theta = 0$  or  $\tan \theta = 1$   
 $\theta = \frac{\pi}{2}, \frac{3\pi}{2}, \frac{\pi}{4}$  or  $\frac{5\pi}{4}$

b) Let  $\sin x + \cos x = R \sin(x+\alpha)$   
 (1)  $\sin x + \cos x = R \sin x \cos \alpha + R \cos x \sin \alpha$   
 $R \cos \alpha = 1$  (1)  
 $R \sin \alpha = 1$  (2)  
 (2)  $\div$  (1)  $\tan \alpha = 1$   
 $\alpha = \frac{\pi}{4}$   
 (1)<sup>2</sup> + (2)<sup>2</sup>  $R^2 \cos^2 \alpha + R^2 \sin^2 \alpha = 1+1$   
 $R^2 = 2$   
 $R = \sqrt{2}$

$\therefore \sin x + \cos x = \sqrt{2} \sin(x + \frac{\pi}{4})$

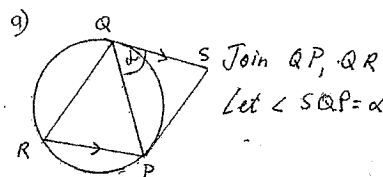


c)  $V = \frac{1}{3} \pi r^2 h$   $r = 3h$   
 (i)  $\therefore V = \frac{1}{3} \pi (3h)^2 \times h$   
 $= 3\pi h^3$   
 at  $t=1$ ,  $h=20$   $\therefore V = 3\pi (20)^3$   
 $= 24000\pi \text{ cm}^3$

(ii)  $\frac{dh}{dt} = \frac{dh}{dV} \times \frac{dV}{dt}$   
 $\frac{dV}{dh} = 9\pi h^2$  at  $t=1$ ,  $h=20$   
 $\therefore \frac{dV}{dh} = 9\pi (20)^2$   
 $= 3600\pi$   
 $\frac{dV}{dt} = 24000\pi$  (from (i))  
 $\therefore \frac{dh}{dt} = \frac{1}{3600\pi} \times 24000\pi$   
 $= \frac{20}{3}$

$\therefore$  height is rising at  $\frac{20}{3}$  cm/minute

QUESTION 3



$\angle QRP = \angle PQS = \alpha$  ( $\angle$  between a tangent and a chord =  $\angle$  in alt. segment)  
 $\angle RPQ = \angle SQP = \alpha$  (alt.  $\angle^s$ ,  $RP \parallel QS$ )  
 $\therefore \angle QRP = \angle QPR = \alpha$   
 $\therefore \Delta QRP$  is isosceles  
 $\therefore QR = QP$  (sides opp.  $= \angle^s$  in isosc.  $\Delta$ )

b) (i) If  $2x+1$  is a factor,  $P(-\frac{1}{2}) = 0$   
 $2(-\frac{1}{2})^3 - (-\frac{1}{2})^2 + K(-\frac{1}{2}) + 1 = 0$   
 $-\frac{1}{4} - \frac{1}{4} - \frac{K}{2} + 1 = 0$   
 $\therefore K = 1$

(ii)  $\therefore P(x) = (2x+1) \cdot Q(x)$   
 $= (2x+1)(x^2-x+1)$   
 For  $x^2-x+1$  to have a root,  $\Delta \geq 0$   
 $\Delta = (-1)^2 - 4 \times 1 \times 1$   
 $= -3$   
 $< 0$   $\therefore x^2-x+1$  has no roots  
 So  $x = -\frac{1}{2}$  is the only root.

c) (i) If  $T = P + Ae^{kt}$   
 $\frac{dT}{dt} = A \times k e^{kt}$   
 $= k \times Ae^{kt}$   
 $= k \times (T - P)$   
 $\therefore T = P + Ae^{kt}$

c) continued

(ii)  $P = 20$ ,  $t=0$ ,  $T=40$  +  $t=1$ ,  $T=30$   
 $T = 20 + Ae^{kt}$   
 $40 = 20 + Ae^0 \Rightarrow A = 20$   
 $\therefore T = 20 + 20e^{kt}$   
 $30 = 20 + 20e^{k}$   
 $e^k = \frac{10}{20} = 0.5$   
 $k = \ln 0.5$   
 $\therefore T = 20 + 20e^{\ln 0.5 \times t}$   
 $t = 3$ ,  $T = 20 + 20e^{\ln 0.5 \times 3}$   
 $= 22.5^\circ$   
 $\therefore$  temperature is  $22.5^\circ$  after 3 hours

#### QUESTION 4

1)  $T_5$  has  $x^4$  term  
 coeff of  $x^4 = {}^nC_4 \cdot 3^{n-4} \times 1^4$   
 coeff of  $x^3 = {}^nC_3 \cdot 3^{n-3} \times 1^3$   
 coeff of  $x^4 = 2 \times$  coeff of  $x^3$   
 $\therefore \frac{n(n-1)(n-2)(n-3)}{4 \times 3 \times 2 \times 1} \times 3^{n-4}$   
 $= \frac{2 \times n(n-1)(n-2)}{3 \times 2 \times 1} \times 3^{n-3}$   
 $\therefore \frac{(n-3)}{4} \times 3^{-1} = 2$   
 $\frac{(n-3)}{4} \times \frac{1}{3} = 2$   
 $n-3 = 24$   
 $n = 27$

2)  $x = 20t \times \cos 30^\circ$   
 $x = 20t \times \frac{\sqrt{3}}{2}$   
 $= 10\sqrt{3}t$   
 $\dot{x} = 10\sqrt{3}$

$y = 20t \times \sin 30^\circ - 5t^2 + 175$   
 $y = 10t - 5t^2 + 175$

(i) stone hits water at  $y=0$   
 $5t^2 - 10t - 175 = 0$   
 $t^2 - 2t - 35 = 0$   
 $(t-7)(t+5) = 0$   
 $t-7=0$  or  $t+5=0$   
 $\therefore t=7$  or  $-5$

but  $t > 0 \therefore t = 7$   
 $\therefore$  It takes 7 seconds to hit the water.

(ii) at  $t=7$ ,  $\dot{x} = 10\sqrt{3}$

$\dot{y} = 10 - 10t$   
 $\therefore \dot{y} = 10 - 10 \times 7$   
 $= -60$

speed  $= \sqrt{(10\sqrt{3})^2 + (-60)^2}$   
 $= 62.4 \text{ m/s}$



c)  $a = \frac{d}{dx} \left( \frac{1}{2} v^2 \right) = 10x - 2x^3$

(i)  $\frac{1}{2} v^2 = 5x^2 - \frac{x^4}{2} + C$

$v=0$ ,  $x=1$

$\therefore 0 = 5 - \frac{1}{2} + C$

$\therefore \frac{1}{2} v^2 = 5x^2 - \frac{x^4}{2} - 4\frac{1}{2}$

$v^2 = 10x^2 - x^4 - 9$

(ii) For the particle to reach the origin,  
 $x=0$

$\therefore v^2 = -9$  which is impossible

$\therefore$  The particle can never reach the origin.

#### QUESTION 5

a) (i)  $x^2 = 8y \Rightarrow y = \frac{x^2}{8}$

$\frac{dy}{dx} = \frac{2x}{8}$   
 $= \frac{x}{4}$

at  $P(4p, 2p^2)$ , m of tangent  $= \frac{4p}{4}$   
 $= p$

$\therefore$  m of normal is  $-\frac{1}{p}$  since

$m_1 m_2 = -1$  for perpendicular lines.

eq'n of normal:  $y - 2p^2 = -\frac{1}{p}(x - 4p)$

$py - 2p^3 = -x + 4p$

$\therefore x + py = 2p^3 + 4p$  is the

eq'n of the normal.

(ii) at  $R$ ,  $x=0$

$\therefore 0 + py = 2p^3 + 4p$

$y = 2p^2 + 4$

$\therefore R(0, 2p^2 + 4)$

(iii)  $M\left(\frac{4p^3}{2}, \frac{2p^3 + 2p^2 + 4}{2}\right)$

$\therefore m(2p, 2p^2 + 2)$

$x=2p$  into  $y = 2p^2 + 2$

$y = 2\left(\frac{x}{2}\right)^2 + 2$

$\therefore$  eq'n of locus is  $y = \frac{x^2}{2} + 2$

b) Show that the result is true for  $n=1$

LHS  $= 1n!$

$= 1$

RHS  $= (1+1)! - 1$

$= 2 - 1$

$= 1$

$=$  LHS  $\therefore$  result is true for  $n=1$

Assume that the result is true for  $n=k$   
 i.e. assume  $1 \times 1! + 2 \times 2! + \dots + k \times k! = (k+1)! - 1$

Using this assumption, show that the result is true for  $n=k+1$ .

i.e. show  $1 \times 1! + 2 \times 2! + \dots + k \times k! + (k+1)(k+1)!$

$= (k+2)! - 1$

is true

LHS  $= (k+1)! - 1 + (k+1)(k+1)!$

$= (k+1)!(1 + k+1) - 1$

$= (k+1)!(k+2) - 1$

$= (k+2)! - 1$

$=$  RHS

$\therefore$  if the result is true for  $n=k$  then it is also true for  $n=k+1$

By the process of mathematical induction, this result is true for all integers  $n=1, 2, 3, \dots$

c)  $P(x) = x^3 - 3px^2 + 2qx - r = 0$

let roots be  $\alpha-d, \alpha, \alpha+d$ .

sum of roots:  $\alpha-d + \alpha + \alpha+d = 3p$

$3\alpha = 3p$

$\alpha = p$  (1)

sum of product of roots, 2 at a time:

$\alpha(\alpha-d) + \alpha(\alpha+d) + (\alpha-d)(\alpha+d) = 2q$

$\alpha^2 - \alpha d + \alpha^2 + \alpha d + \alpha^2 - d^2 = 2q$

$3\alpha^2 - d^2 = 2q$

but  $\alpha = p \therefore 3p^2 - d^2 = 2q$

$d^2 = 3p^2 - 2q$

product of roots:  $(\alpha-d) \cdot \alpha \cdot (\alpha+d) = r$

$\alpha(\alpha^2 - d^2) = r$

$\alpha^3 - \alpha d^2 = r$

$p^3 - p(3p^2 - 2q) = r$

$\therefore r = p^3 - 3p^3 + 2pq$

$= -2p^3 + 2pq$

$r = 2p(q - p^2)$

or from (1)  $\alpha = p$  is a soln so

$P(\alpha) = 0$  and  $P(p) = 0$

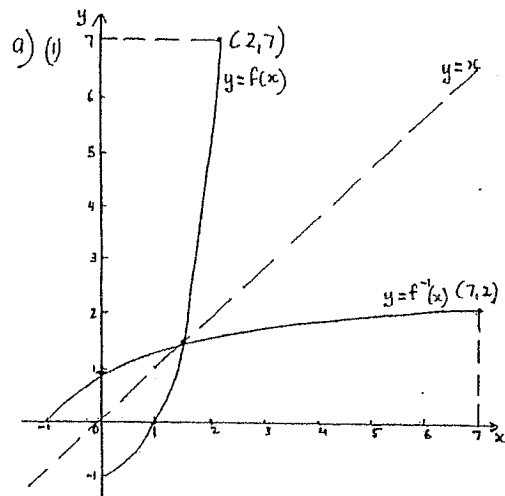
$\therefore p^3 - 3p^3 + 2qp - r = 0$

$-2p^3 + 2qp - r = 0$

$r = 2qp - 2p^3$

$r = 2p(q - p^2)$

# QUESTION 6



b)  $(1+x)^n = 1 + {}^nC_1 x + {}^nC_2 x^2 + \dots + {}^nC_{n-1} x^{n-1} + {}^nC_n x^n$   
 Differentiating both sides with respect to  $x$ :  
 $n(1+x)^{n-1} = {}^nC_1 + 2 {}^nC_2 x + 3 {}^nC_3 x^2 + \dots + (n-1) {}^nC_{n-1} x^{n-2} + n {}^nC_n x^{n-1}$   
 Let  $x=1$ ,  
 $n \cdot 2^{n-1} = {}^nC_1 + 2 {}^nC_2 + 3 {}^nC_3 + \dots + (n-1) {}^nC_{n-1} + n$   
 ${}^nC_1 + 2 {}^nC_2 + 3 {}^nC_3 + \dots + (n-1) {}^nC_{n-1} = n \cdot 2^{n-1} - n$   
 $\therefore ({}^nC_1) + 2({}^nC_2) + 3({}^nC_3) + \dots + (n-1)({}^nC_{n-1}) = n(2^{n-1} - 1)$

(iii) Any function and its inverse must intersect in the line  $y=x$  since this is their axis of symmetry.  
 Pnt of intersection of  $y=x^3-1$  and  $y=x$   
 $x^3-1=x \therefore x^3-x-1=0$

(iv) If  $x=1$ ,  $g(x) = 1-1-1 = -1 < 0$

If  $x=2$ ,  $g(x) = 8-2-1 = 5 > 0$

$g(x)$  changes sign  $\therefore$  there is a root between  $x=1$  and  $x=2$

(v)  $g'(x) = 3x^2-1$

Let  $x_1 = 1.5$

$$x_2 = x_1 - \frac{g(x_1)}{g'(x_1)}$$

$$= 1.5 - \frac{(1.5)^3 - 1.5 - 1}{3(1.5)^2 - 1}$$

$$= 1.3478\dots$$

$$\approx 1.35$$

$\therefore$  a better approximation to the root is  $x=1.35$

# QUESTION 7

a) let  $m = n + \frac{1}{n}$

$$\therefore m^2 - 5m + 6 = 0$$

$$(m-3)(m-2) = 0$$

$$m = 3 \text{ or } 2$$

$$\therefore n + \frac{1}{n} = 3 \quad \text{or} \quad n + \frac{1}{n} = 2$$

$$n^2 - 3n + 1 = 0 \quad \text{or} \quad n^2 - 2n + 1 = 0$$

$$n = \frac{3 \pm \sqrt{9-4}}{2} \quad (n-1)^2 = 0$$

$$\therefore n = \frac{3 \pm \sqrt{5}}{2} \quad \text{or } 1$$

b) (i)  $\frac{dy}{dx} = \frac{(e^x + e^{-x})(e^x + e^{-x}) - (e^x - e^{-x})(e^x - e^{-x})}{(e^x + e^{-x})^2}$   
 $= \frac{e^{2x} + 2e^{-2x} - e^{2x} + 2e^{-2x}}{(e^x + e^{-x})^2}$   
 $= \frac{4}{(e^x + e^{-x})^2}$

$$\neq 0 \text{ for any value of } x$$

$\therefore$  The function has no stationary points.

(ii) If  $y=1$ , then  $\frac{e^x - e^{-x}}{e^x + e^{-x}} = 1$

$$e^x - e^{-x} = e^x + e^{-x}$$

$$\therefore 2e^{-x} = 0$$

$$e^{-x} = 0 \text{ which is}$$

not possible  $\therefore y \neq 1$

If  $y=-1$  then  $\frac{e^x - e^{-x}}{e^x + e^{-x}} = -1$

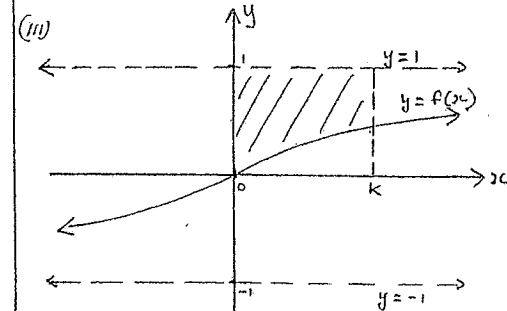
$$e^x - e^{-x} = -e^x - e^{-x}$$

$$2e^x = 0$$

$$\therefore e^x = 0 \text{ which is}$$

impossible.

$\therefore y = \pm 1$  are asymptotes



(iv) Area =  $k - \int_0^k \frac{e^x - e^{-x}}{e^x + e^{-x}} dx$   
 $= k - \left[ \ln(e^x + e^{-x}) \right]_0^k$   
 $= k - \ln(e^k + e^{-k}) + \ln(e^0 + e^0)$   
 $= k - \ln(e^k + e^{-k}) + \ln 2$

(v) Area =  $k \times \ln e - \ln(e^k + e^{-k}) + \ln 2$   
 $= \ln e^k - \ln(e^k + e^{-k}) + \ln 2$   
 $= \ln \frac{e^k}{e^k + e^{-k}} + \ln 2$

since  $e^k$  and  $e^{-k}$  are both  $> 0$  for all  $k$ ,

$$\frac{e^k}{e^k + e^{-k}} < 1 \text{ for all } k$$

$$\therefore \ln \frac{e^k}{e^k + e^{-k}} < 0$$

$$\therefore \text{Area} < \ln 2 \text{ for all values of } k$$